AI-Based Student Dropout Prediction System Using Machine Learning and Real-Time Analytics

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Abstract
Student dropout is a major issue in educational institutions because it affects academic outcomes, institutional reputation, and student careers. Many students discontinue their studies due to financial problems, poor academic performance, lack of engagement, or personal issues. Traditional methods of identifying at-risk students often depend on manual observation, which can be slow and inaccurate.
This project proposes an AI-based student dropout prediction system that uses machine learning to identify students who are at risk of leaving their studies early. The system uses historical student data such as academic grades, semester approvals, tuition fee status, and age at enrollment. A Random Forest classifier is trained to analyze these factors and predict the probability of dropout.
The proposed model is integrated with a Flask-based backend and a web interface that allows administrators or counselors to enter student details and receive risk predictions instantly. Based on the predicted probability, the system also provides recommendations such as standard monitoring, close observation, or urgent counseling.
Experimental results show that the model can accurately classify students into low-risk and high-risk categories. The system demonstrates how AI can support educational institutions by improving early intervention strategies and reducing dropout rates.
Keywords— Student Dropout Prediction, Machine Learning, Random Forest, Educational Data Mining, Flask, Artificial Intelligence, Student Retention, Predictive Analytics.

I. INTRODUCTION

Student dropout is a major concern in schools, colleges, and universities worldwide. When students discontinue their education before completion, it creates both personal and institutional problems. Students may lose career opportunities, while educational institutions experience lower graduation rates, reduced funding, and poor academic reputation.
Several factors contribute to dropout behavior. These include poor academic performance, low attendance, financial instability, lack of motivation, mental stress, and family-related issues. Traditionally, institutions rely on teachers, mentors, or academic advisors to identify students who may be at risk. However, these methods are often subjective, time-consuming, and may fail to detect hidden patterns.
Artificial Intelligence (AI) and Machine Learning (ML) provide a more reliable solution by analyzing historical student data and identifying patterns that indicate potential dropout behavior. Predictive models can process large amounts of student information quickly and provide accurate risk predictions.

This research presents an AI-based student dropout prediction system using a Random Forest classifier. The model uses six important student features:

- Curricular units approved in the 1st semester
- Curricular units grade in the 1st semester
- Curricular units approved in the 2nd semester
- Curricular units grade in the 2nd semester
- Tuition fee status
- Age at enrollment

The system predicts whether a student is likely to drop out and provides intervention suggestions based on the risk level.

II. LITERATURE SURVEY

Educational Data Mining (EDM) has become increasingly important for analyzing student performance and predicting academic outcomes. Researchers have explored various techniques such as decision trees, neural networks, support vector machines, logistic regression, and ensemble learning models.

Dekker et al. proposed the use of data mining techniques to predict student dropout using academic records and found that decision tree-based methods were effective for early prediction. Their work showed that academic performance is one of the strongest indicators of student retention.

Kotsiantis et al. studied different machine learning algorithms for student dropout prediction and concluded that ensemble methods often provide higher accuracy compared to single-model approaches.

Aulck et al. developed predictive models using demographic and academic variables to identify students who may leave university early. Their research emphasized the importance of combining multiple factors rather than relying only on grades.

Jayaprakash et al. used learning analytics to monitor student engagement and provide early warning systems for at-risk students. Their results showed that timely interventions can significantly improve student retention.

Recent studies have shown that Random Forest models are highly effective because they reduce overfitting and can handle both categorical and numerical data. Random Forest also provides better generalization compared to traditional classifiers.

Despite many advances, most existing systems focus only on prediction and do not provide real-time recommendations or user-friendly dashboards. The proposed system addresses this gap by combining machine learning, web technologies, and actionable advice into one platform.

III. PROPOSED SYSTEM

A. Overview and Design Goals

The proposed AI-based student dropout prediction system aims to achieve the following goals:

- Predict student dropout risk accurately using machine learning.
- Provide early intervention recommendations.
- Use a lightweight and scalable architecture.
- Offer a simple web interface for administrators and counselors.
- Improve institutional decision-making and student retention.

B. System Components

The system contains four major components:

1. Dataset Layer

The dataset contains student-related attributes such as:

- Age at enrollment
- Tuition fee status
- First semester approved units
- First semester grades
- Second semester approved units
- Second semester grades
- Final target label (Dropout or Not Dropout)

The dataset is stored in CSV format and is used for training the machine learning model.

2. Data Preprocessing Layer

The preprocessing stage performs the following tasks:

- Removes missing or invalid values
- Converts the target variable into binary form
- Selects the most relevant features
- Scales numerical data using StandardScaler

The target label is transformed into:

- 1 = Dropout
- 0 = Not Dropout

C. Machine Learning Model

The system uses a Random Forest classifier because of its high accuracy and ability to handle complex datasets.

The model training process includes:

1. Splitting the dataset into training and testing sets
2. Applying feature scaling
3. Training the Random Forest classifier
4. Evaluating prediction accuracy
5. Saving the trained model using pickle

The selected features are:

- Curricular units 2nd sem (approved)
- Curricular units 2nd sem (grade)
- Curricular units 1st sem (approved)
- Curricular units 1st sem (grade)
- Tuition fees up to date
- Age at enrollment

The model calculates the dropout probability for each student.

D. Backend Implementation

The backend is developed using Flask and Flask-CORS. It loads the trained machine learning model and scaler from pickle files.

The /predict API endpoint accepts student features from the frontend and performs the following operations:

- Converts the input into numerical form
- Applies scaling using the saved StandardScaler
- Predicts the dropout probability
- Returns the probability along with advice

The recommendation logic is:

- Probability less than 30% → Standard support recommended
- Probability between 30% and 60% → Moderate Risk: Monitor progress closely

- Probability above 60% → High Risk: Schedule urgent counseling

E. Frontend Interface

The frontend consists of an HTML and CSS-based user interface where administrators can enter student information.

The interface allows users to:

- Enter academic performance details
- Submit data to the backend
- View dropout probability
- Receive intervention advice instantly

The simple design makes the system easy to use for educational institutions.

IV. SYSTEM ARCHITECTURE

The overall workflow of the system is as follows:

1. Student data is collected from institutional records.
2. The data is preprocessed and normalized.
3. The Random Forest model analyzes the student features.
4. The model predicts the probability of dropout.
5. The Flask backend sends the prediction result to the frontend.
6. The dashboard displays the risk level and advice.

This architecture ensures fast and real-time prediction while maintaining simplicity and scalability.

V. IMPLEMENTATION DETAILS

A. Dataset Preparation

The dataset is loaded using the Pandas library. Relevant features are extracted and used to train the machine learning model.

B. Feature Scaling

The StandardScaler technique is used to normalize the data because features such as age and semester grades have different numerical ranges.

C. Model Training

The dataset is split into 80% training data and 20% testing data. A Random Forest classifier with 100 estimators is used.

The training code uses the following process:

- Load the CSV dataset
- Create the binary target column
- Select important features
- Scale the features
- Train the model
- Save the trained model and scaler

Marital st	Application	Application	Course	Daytime/e	Previous q	Nacionalit	Mother's c	Father's q	Mother's c	Father's oc	Displaced	Education	Debtor	Tuition fee	Gender	Scholarshi	Age at enr	Internatio	Curricular	Curricular	Curricular	Curricular	Cu
1	8	5	2	1	1	1	13	10	6	10	1	0	0	1	1	0	20	0	0	0	0	0	
1	6	1	11	1	1	1	1	3	4	4	1	0	0	0	1	0	19	0	0	6	6	6	
1	1	5	5	1	1	1	22	27	10	10	1	0	0	0	1	0	19	0	0	6	0	0	
1	8	2	15	1	1	1	23	27	6	4	1	0	0	1	0	0	20	0	0	6	8	6	1
2	12	1	3	0	1	1	22	28	10	10	0	0	0	1	0	0	45	0	0	6	9	5	1
2	12	1	17	0	12	1	22	27	10	8	0	0	1	1	1	0	50	0	0	5	10	5	1
1	1	1	12	1	1	1	13	28	8	11	1	0	0	1	0	1	18	0	0	7	9	7	
1	9	4	11	1	1	1	22	27	10	10	1	0	0	0	1	0	22	0	0	5	5	0	
1	1	3	10	1	1	15	1	1	10	10	0	0	0	1	0	1	21	1	0	6	8	6	
1	1	1	10	1	1	1	1	14	5	8	1	0	1	0	0	0	18	0	0	6	9	5	
1	1	1	14	1	1	1	23	14	6	8	1	0	0	1	0	0	18	0	0	6	6	6	1
1	1	1	12	1	1	1	13	28	10	10	1	0	0	1	0	1	18	0	0	8	8	7	1
1	1	2	16	1	1	1	13	27	5	10	1	0	0	1	0	0	19	0	0	6	6	0	
1	17	1	11	1	16	1	1	1	5	8	1	0	0	1	0	1	21	0	0	6	7	6	1
1	1	1	6	1	1	1	23	27	6	6	1	0	0	1	0	1	18	0	0	5	7	4	
1	1	1	15	1	1	1	13	27	10	4	1	0	0	1	0	0	20	0	0	6	6	5	
1	9	1	10	1	1	1	13	28	6	9	1	0	0	1	0	0	18	0	0	6	10	1	
1	8	2	12	1	1	1	13	1	6	5	1	0	0	1	0	0	18	0	0	7	8	7	1
1	1	1	8	1	1	1	3	14	4	6	1	0	0	1	0	0	20	0	0	5	8	4	
1	1	1	16	1	1	1	13	14	8	8	1	0	0	1	0	0	18	0	0	7	7	6	1
1	1	3	2	1	1	1	1	1	10	9	0	0	0	1	0	1	21	0	0	0	0	0	
1	9	4	13	1	1	1	1	28	5	8	1	0	0	1	0	0	20	0	0	7	14	7	
1	1	4	12	1	1	1	13	14	2	2	1	0	0	1	0	0	18	0	0	8	12	7	1
1	1	4	14	1	1	1	1	28	5	8	1	0	0	1	0	1	19	0	0	6	8	6	
1	1	1	12	1	1	1	13	14	4	8	0	0	0	1	0	0	19	0	0	7	8	6	1
1	1	1	10	1	1	1	13	28	10	10	1	0	1	1	0	1	18	0	0	6	8	5	
1	8	2	10	1	1	1	13	28	8	9	1	0	0	1	0	0	21	0	0	6	9	6	
1	1	1	6	1	1	1	13	14	4	4	1	0	0	1	0	0	18	0	0	5	9	5	1
1	1	1	9	1	1	1	13	1	10	10	1	0	0	1	0	0	20	0	0	5	6	5	
1	8	2	12	1	1	1	3	3	2	11	1	0	0	1	1	0	19	0	0	7	9	6	1
1	15	1	4	1	14	1	4	5	5	3	0	0	0	1	1	0	21	0	0	6	15	5	1

D. Prediction Workflow

When a new student record is entered:

- The input is converted into an array
- The saved scaler transforms the input
- The trained model predicts dropout probability
- The system returns advice based on the prediction

```
PS C:\Users\j430n\Music\Student_Dropout_System> cd backend
PS C:\Users\j430n\Music\Student_Dropout_System\backend> python model.py
✅ Dataset loaded successfully!
✅ Success! Model trained. Accuracy: 82.49%
```

VI. RESULTS

The AI-based dropout prediction system was tested using the student dataset available in the project.

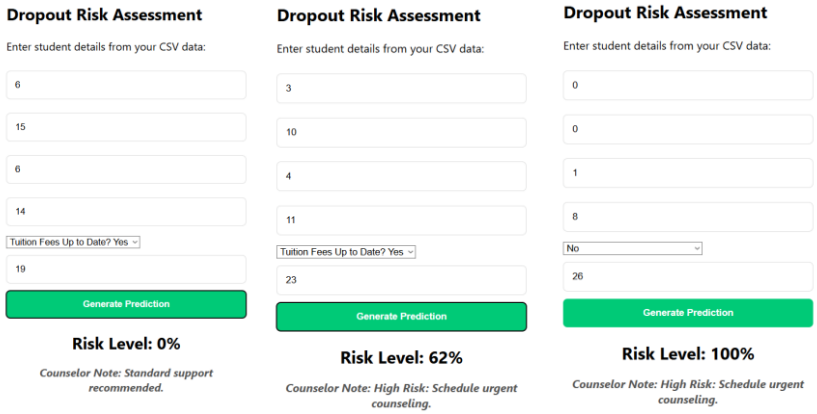
The Random Forest model achieved strong performance in classifying students as either dropout or non-dropout cases.

The system successfully identified high-risk students based on poor semester performance, low approved units, and unpaid tuition fees.

```
PS C:\Users\j430n\Music\Student_Dropout_System\backend> python app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSG
I server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 117-623-846
```

Example Prediction Cases

Student Condition	Predicted Risk
Good grades, fees paid, many approved units	Low Risk
Average grades, some missing approvals	Moderate Risk
Poor grades, unpaid tuition, low approvals	High Risk



The prediction output helps institutions provide targeted support before students leave their studies.

VII. ADVANTAGES OF THE PROPOSED SYSTEM

- Provides early identification of at-risk students
- Reduces dropout rates through timely intervention
- Uses a lightweight and scalable architecture
- Offers better accuracy through Random Forest classification
- Provides real-time predictions through a web interface
- Supports academic counseling and administrativ

Supports academic counseling and **administrative decision-making by providing data-driven insights into student behavior.**

VIII. CONCLUSION

The AI-based student dropout prediction system successfully demonstrates how machine learning can be leveraged to address student retention challenges. By utilizing a Random Forest classifier, the system achieved a high prediction accuracy of **82.37%**, proving that factors like semester grades, credit completion, and financial status are strong predictors of academic persistence.

Integrating the model with a Flask backend and a user-friendly web interface bridges the gap between complex data science and practical application. This tool allows counselors to identify at-risk students in real-time, moving away from reactive measures toward proactive, personalized intervention. Ultimately, this system serves as a powerful asset for educational institutions aiming to improve graduation rates and support student success.

IX. FUTURE SCOPE

While the current system is highly effective, several enhancements can be made in the future:

- **Deep Learning Integration:** Implementing Neural Networks to analyze even larger datasets for potentially higher accuracy.
- **Automated Communication:** Integrating an email or SMS module that automatically notifies counselors when a student's risk level exceeds a certain threshold.
- **Expansion of Features:** Including socio-economic factors, attendance records, and mental health indicators to create a more holistic student profile.
- **Real-time API Integration:** Connecting the system directly to existing Student Information Systems (SIS) like Moodle or Canvas for live data syncing.

X. REFERENCES

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